

Birla Institute of Technology & Science, Pilani, Rajasthan
First Semester 2021-2022
Comprehensive Exam-Part B (Open Book)

EEE F311 Communication Systems

Date: 27-12-2021, Duration: 120 Minutes, Maximum Marks: 40

- There are 4 questions each of 10 marks.
- Attempt the parts of a question sequentially.
- Enclose the final answer in a box.

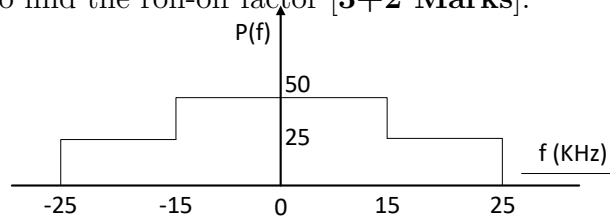
1. The generator matrix of a (7, 4) linear block code is given as

$$\mathbf{G} = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

- (a) Find all the code word, and the error detecting capability. [4+1 Marks].
- (b) Decode the transmitted data vectors for the following received data vectors $\mathbf{r}_1 = 1110001$, $\mathbf{r}_2 = 1001100$, $\mathbf{r}_3 = 0101011$, $\mathbf{r}_4 = 1100111$, and $\mathbf{r}_5 = 1010101$ [5 Marks].
2. (a) A binary communication channel has the following transition matrix. Find the channel capacity if input symbols are equi-probable [5 Marks].

$$\begin{bmatrix} 2/3 & 1/3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- (b) The frequency domain of a pulse is given in the figure below. Find $p(t)$ and show that it is a Nyquist pulse. Also find the roll-off factor [3+2 Marks].



3. A binary signaling scheme uses $q(t)$ to transmit bit 0 and $p(t)$ to transmit bit 1 with unequal probabilities $\frac{1}{3}$ and $\frac{2}{3}$, respectively. Consider AWGN channel with $\frac{N_0}{2} = 1$. The pulses are given as

$$q(t) = \begin{cases} 2, & 0 \leq t \leq 5 \\ -1, & 5 < t \leq 10 \\ 0, & \text{otherwise} \end{cases} \quad p(t) = \begin{cases} 2, & 5 \leq t \leq 10 \\ 0, & \text{otherwise} \end{cases}$$

- (a) Derive an expression for the optimum threshold and find its numerical value upto two decimal places [5 Marks].
 - (b) Find the average bit error rate in terms of Q -function [3 Marks].
 - (c) Draw the basis signal representation for the signals $q(t)$ and $p(t)$. [2 Marks].
4. An analog signal $m(t)$ is replicated through a discrete signal $m[n]$ using a digital computer sampled at the Nyquist sampling rate, as shown in Figure 1. The duration of the signal is 1 second, and the average power of the signal is $P_m = 1.3$ Watt.

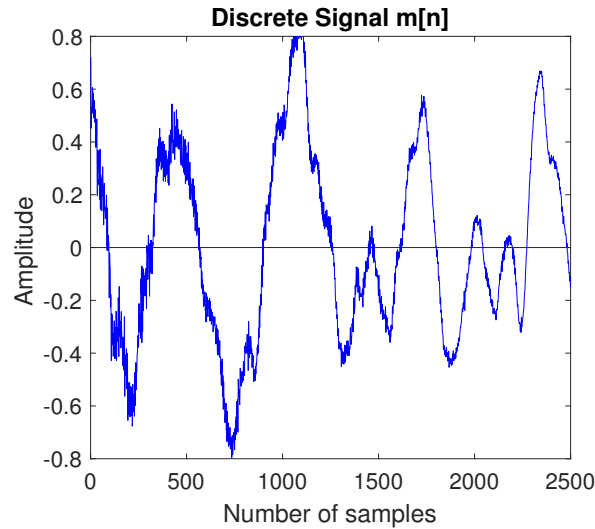


Figure 1: Message Signal

- (a) You are given the task to design an analog transceiver to transmit message $m(t)$ of Figure 1 over a band-limited channel. Here, the analog signal $m(t)$ should be a FM modulated signal using a carrier signal of 88 KHz with a transmit power of 50 Watt, and the frequency deviation $\Delta_f = 64$ Hz [5 Marks].
 - i. Design the FM transmitter if the initial frequency sensitivity coefficient is $k_f = 10$ volt/Hz.
 - ii. Determine the channel bandwidth to transmit the FM signal.
 - iii. Draw a detector block diagram to receive the message signal.
- (b) You are given the task to design a digital baseband transceiver to transmit $m(t)$ of Figure 1 over a band-limited channel [5 Marks].
 - i. Determine the specification of analog-to-digital converter (ADC) using PCM to achieve an SQNR of 50 dB.
 - ii. The PCM binary digits to be line encoded and transmitted using Nyquist pulses at a symbol rate of 25 symbols per second over an AWGN channel with a channel bandwidth of 25 Hz. Find the impulse response of the Nyquist pulse.
 - iii. Find the minimum time taken (with minimal probability of error) to transmit the whole data for an AWGN channel with Gaussian signaling with mean square power for the signal as $P_s = 1.5$ Watt and Noise PSD $N_0/2 = 2 \times 10^{-3}$ W/Hz.